

Web Engineering & Development (SWE 363)

Back-end Development Fundamentals

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In today's lecture:

- Express.js Request data
 - Query String Parameters
 - Route Parameters
 - Param Middleware
 - Request Body
 - Sending JSON Responses
 - Status Codes

Reference:

- Zybook: 6.5

6.5 Express request data

How do we get data from the client?

When a client makes a request to our server, they often send **data**:

- **Query parameters:** `/search?q=react&page=2`
- **Route parameters:** `/users/123` or `/profile/John/Doe`
- **Request body:** JSON data, form data, etc.

Today we'll learn how to **read** this data in Express!

Three ways to send data

Method	Location	Example	Use Case
Query String	URL	/search? q=react&page=2	Optional filters, search terms
Route Parameters	URL path	/users/123	Identifying specific resources
Request Body	HTTP body	POST with JSON	Creating/updating data

1. Query String Parameters

What are Query String Parameters?

Query strings are key-value pairs added to URLs after `?`

```
http://localhost:3000/search?q=react&page=2&sort=date
```

Format: `?key1=value1&key2=value2&key3=value3`

Why use Query Strings?

Optional – can be added or removed easily

Bookmarkable – URLs can be saved/shared

Cacheable – browsers can cache different query combinations

Flexible – don't change the resource path

Common uses: Search terms, filters, pagination, sorting

Reading Query Parameters in Express

Use `req.query` to access query string parameters

```
app.get("/search", (req, res) => {  
  const { q, page, sort } = req.query;  
  // q = "react"  
  // page = "2"  
  // sort = "date"  
  
  res.json({ query: q, page, sort });  
});
```

Important: Values are always **strings**!

Query Parameters: Important Notes

Values are strings - convert if needed:

```
const page = Number(req.query.page) || 1;
```

Always validate - check if required params exist:

```
if (!req.query.name) {  
  return res.status(400).json({ error: "name required" });  
}
```

Optional by nature - missing params are `undefined`

2. Route Parameters

What are Route Parameters?

Route parameters are **named placeholders** in the URL path

```
app.get("/users/:userId", (req, res) => {  
  // :userId is a route parameter  
});
```

URL: /users/123

Parameter: userId = "123"

They identify **specific resources** in the path!

Why use Route Parameters?

Clean URLs - `/users/123` vs `/users?id=123`

RESTful - follows REST API conventions

Descriptive - makes the resource clear

Hierarchical - supports nested resources

Example: `/orders/987/items/2` (order 987, item 2)

Reading Route Parameters

Use `req.params` to access route parameters

```
app.get("/users/:userId", (req, res) => {  
  const { userId } = req.params;  
  // userId = "123" (string)  
  
  res.json({ userId });  
});
```

URL: `/users/123` → `userId = "123"`

Multiple Route Parameters

You can have multiple parameters in one route:

```
app.get("/profile/:first/:last", (req, res) => {  
  const { first, last } = req.params;  
  
  res.json({  
    ok: true,  
    fullName: `${first} ${last}`  
  });  
});
```

URL: /profile/Jack/Black

Result: { ok: true, fullName: "Jack Black" }

Route Parameters: Important Notes

Values are strings - convert to numbers if needed:

```
const userId = Number(req.params.userId);
```

Required by route - if missing, route won't match

Order matters - `/users/:id/posts/:postId` vs `/users/:postId/posts/:id`

Query vs Route Parameters

Feature	Query Parameters	Route Parameters
Syntax	?key=value	/:paramName
Required?	Optional	Required (for route match)
Use Case	Filters, search, pagination	Resource identification
Example	/search?q=react	/users/123
Access	req.query	req.params

3. Route Parameter Middleware

The Problem: Repeated Validation

What if multiple routes use the same parameter? **Problem:** Code duplication!

```
app.get("/users/:userId", (req, res) => {
  const userId = Number(req.params.userId);
  if (!Number.isFinite(userId) || userId < 1) {
    return res.status(400).json({ error: "Invalid userId" });
  }
  // ... use userId
});

app.delete("/users/:userId", (req, res) => {
  const userId = Number(req.params.userId);
  if (!Number.isFinite(userId) || userId < 1) {
    return res.status(400).json({ error: "Invalid userId" });
  }
  // ... same validation again!
});
```

Solution: Param Middleware

`app.param()` runs **once** when a route with that parameter matches

```
app.param("userId", (req, res, next, userId) => {
  const num = Number(userId);
  if (!Number.isFinite(num) || num < 1) {
    return res.status(400).json({
      ok: false,
      error: "userId must be positive number"
    });
  }
  req.userIdNum = num; // Store validated value
  next(); // Continue to route handler
});
```

Runs automatically for any route with `:userId` !

Using Param Middleware

After defining param middleware, routes can use the validated value:

```
// Middleware (runs first)
app.param("userId", (req, res, next, userId) => {
  const num = Number(userId);
  if (!Number.isFinite(num) || num < 1) {
    return res.status(400).json({
      ok: false, error: "userId must be positive number"
    });
  }
  req.userIdNum = num;
  next();
});
// Route (uses validated value)
app.get("/users/:userId", (req, res) => {
  res.json({ ok: true, userId: req.userIdNum });
});
```

Param Middleware: How It Works

1. **Request comes in:** `/users/42`
2. **Express matches route:** `GET /users/:userId`
3. **Param middleware runs:** `app.param("userId", ...)`
 - Validates and converts `"42"` → `42`
 - Stores in `req.userIdNum`
 - Calls `next()`
4. **Route handler runs:** Uses `req.userIdNum`

One validation, multiple routes benefit!

Param Middleware: Benefits

DRY – Don't Repeat Yourself

Centralized validation – one place to fix bugs

Consistent – all routes get same validation

Reusable – works for GET, POST, DELETE, etc.

Best for: Type conversion, validation, normalization

4. Receiving JSON Data

Request Body: Another Way to Send Data

So far we've seen data in URLs (query strings, route params)

Request body is data sent in the HTTP request body:

- **JSON:** `{ "name": "Ali", "age": 22 }`
- **Form data:** `name=Ali&age=22`
- **Files:** `multipart/form-data`

Common for **POST** and **PUT** requests

Why Use Request Body?

URLs have limits:

- Query strings get long and messy
- Route params are for resource IDs
- Sensitive data shouldn't be in URLs

Request body is better for:

- Large amounts of data
- Complex objects
- Sensitive information (passwords, etc.)
- Creating/updating resources

Receiving JSON in Express

To receive JSON data, you need **middleware** to parse it:

```
import express from "express";
const app = express();

// Enable JSON parsing middleware
app.use(express.json());

// Now you can read JSON from req.body
app.post("/users", (req, res) => {
  const { name, age } = req.body;
  // name = "Ali"
  // age = 22
  res.json({ ok: true, user: { name, age } });
});
```

Important: `app.use(express.json())` must come **before** your routes!

Receiving URL-Encoded Data

For traditional HTML forms, use `express.urlencoded()` :

```
app.use(express.urlencoded({ extended: true }));

app.post("/login", (req, res) => {
  const { username, password } = req.body;
  // Data comes from form: username=john&password=secret
  // ...
});
```

Difference:

- `express.json()` - parses JSON bodies
- `express.urlencoded()` - parses form data (`key=value&key2=value2`)

Request Body: Important Notes

Always validate - check required fields:

```
if (!req.body.name) {  
  return res.status(400).json({ error: "name required" });  
}
```

Middleware order matters - `app.use(express.json())` before routes

Data types - JSON preserves types (numbers, booleans, etc.)

Security - validate and sanitize all input!

Summary: All Ways to Get Data

Source	Access	Example	Use Case
Query String	req.query	/search?q=react	Optional filters
Route Params	req.params	/users/123	Resource ID
Request Body	req.body	POST with JSON	Creating/updating data

5. Sending JSON Responses

Sending JSON in Express

Use `res.json()` to send JSON responses:

```
app.get("/data", (req, res) => {  
  res.json({ ok: true, message: "Success!" });  
});
```

Automatically:

- Sets `Content-Type: application/json`
- Converts object to JSON string
- Sends response

Setting Status Codes

Use `res.status()` before `res.json()` :

```
// Success (200 is default)
res.json({ ok: true, data: result });

// Error (400 Bad Request)
res.status(400).json({
  ok: false,
  error: "Invalid input"
});

// Not Found (404)
res.status(404).json({
  ok: false,
  error: "Resource not found"
});
```

Common Status Codes

Code	Meaning	Use Case
200	OK	Successful request
400	Bad Request	Invalid input/parameters
404	Not Found	Resource doesn't exist
500	Server Error	Server-side error

Always include status codes for errors!

Summary: Request Data in Express

Type	Access	Example	Use Case
Query	req.query	/search?q=react	Optional filters
Route	req.params	/users/123	Resource ID
Body	req.body	POST with JSON	Creating/updating data
Param Middleware	app.param()	Validates route params	Reusable validation

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Demo

Starter code at: https://classroom.github.com/a/f7Xq_C38